

REINFORCEMENT LEARNING

From Fundamentals to Advanced Policy Methods

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Outline

Java
A.I.M.A. → Artificial Intelligence:
A Modern Approach

1 Introduction to Reinforcement Learning Fundamentals

2 Advanced Reinforcement Learning Algorithms

3 Policy Optimization Methods

4 Advanced Policy Methods & Multi-Agent RL

Russell
& Norvig



Outline

Learning Rate

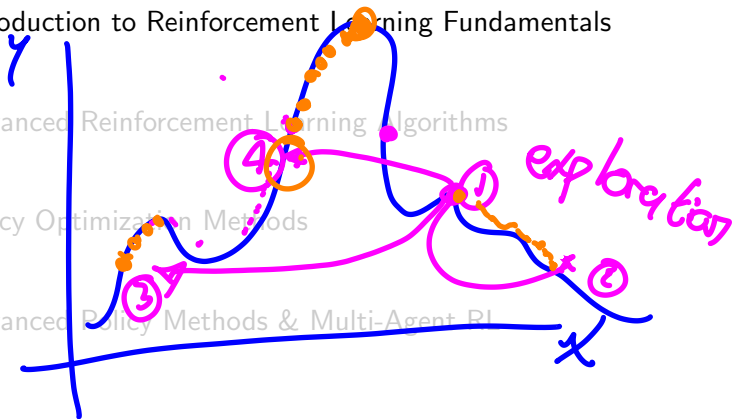
exp/ortation

1 Introduction to Reinforcement Learning Fundamentals

2 Advanced Reinforcement Learning Algorithms

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What is Reinforcement Learning?

Reinforcement Learning

- **Reinforcement Learning** is a computational approach to learning whereby an **agent** tries to maximize the notion of cumulative **reward**.
- The agent learns to achieve a **goal** in an uncertain, potentially complex **environment**.

Mario Bros



time

→ reward
[500-time]
time

score



What is Reinforcement Learning?

Reinforcement Learning

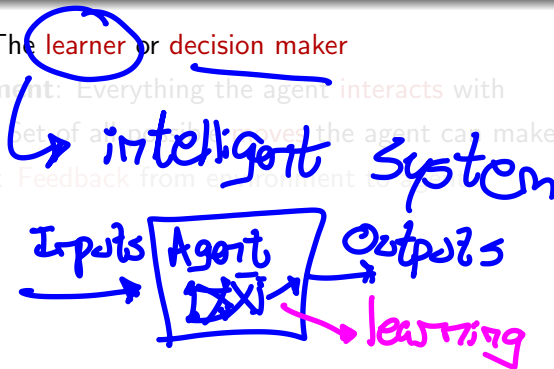
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- **Agent:** The **learner** or **decision maker**

- **Environment:** Everything the agent interacts with

- **Actions:** Set of all possible moves the agent can make

- **Rewards:** Feedback from environment



What is Reinforcement Learning?

Reinforcement Learning

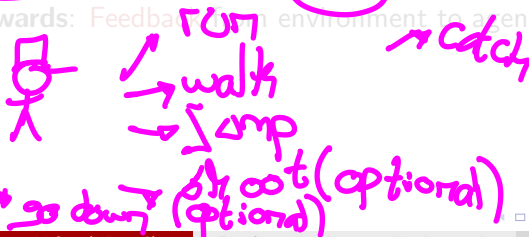
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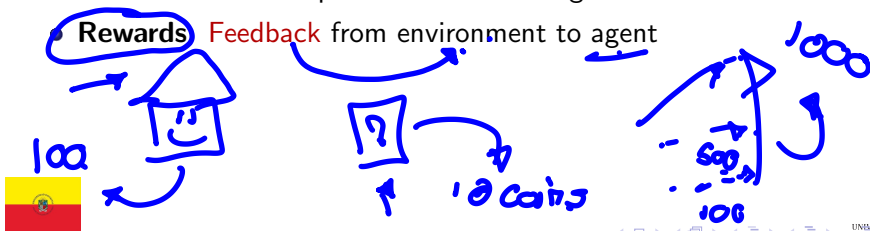
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RL vs. Other Learning Paradigms

Supervised Learning:

- Uses labeled data
- Direct feedback
- Static datasets



Unsupervised Learning:

- No labels
- Pattern discovery
- Structure finding

Handwritten text: "clusters" with an arrow pointing to a red circle containing a cluster of points.

Handwritten equation:
$$z = [a_1 w_1 + a_2 w_2 + a_3 w_3]$$

Handwritten text: "Complex" with a red arrow pointing to a red circle containing a complex network diagram.

Handwritten text: "No - can vex" with a red arrow pointing to a red circle containing a complex network diagram.

Handwritten equation:
$$w_1 + w_2 + w_3 = 1.0$$



RL vs. Other Learning Paradigms

Supervised Learning:

- Uses labeled data
- Direct feedback
- Static datasets

Unsupervised Learning:

- No labels
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Reinforcement Learning:

- Trial-and-error learning
- Delayed rewards
- Sequential decision making
- Exploration vs. exploitation
- Dynamic environments

scientific method

time → simulation

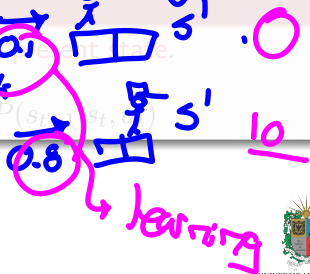
$a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n \Rightarrow \text{Reward}$
 $\downarrow$
 $d_{a1}?$ $d_{a2}?$ $d_{a_n}?$



Markov Decision Processes (MDPs)

An MDP is defined by the tuple (S, A, P, R, γ) :

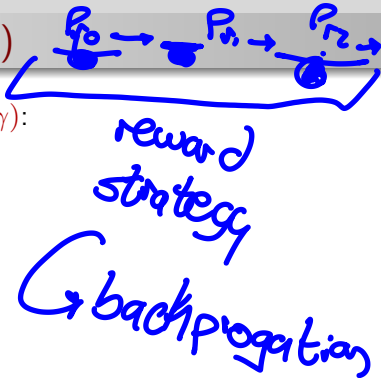
- **S**: Set of **states** $\rightarrow$ **environment**
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- γ : **Discount** factor $[0, 1]$



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Markov Property

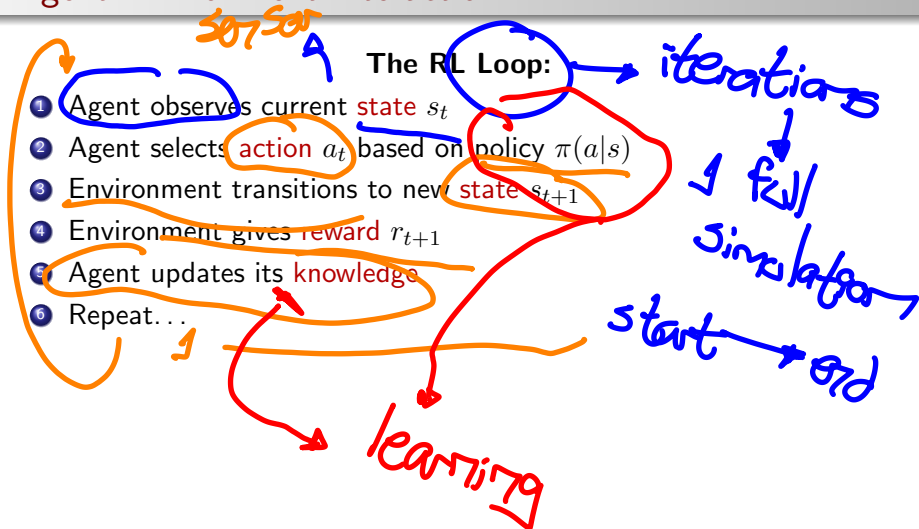
The future is independent of the past given the present state.

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots) = P(s_{t+1} | s_t, a_t)$$

Bayes Theorem $\leftrightarrow$ conditional probability



Agent-Environment Interaction



Value Functions and Policies

Policy

A **policy** $\pi(a|s)$ is a mapping from **states** to **probability distributions** over actions.

$$\pi(a|s) \rightarrow$$

$$a_0 \leftarrow p_j = w_j$$

$$a_1 \leftarrow p_e = w_e$$

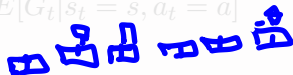
• State Value: $V^\pi(s) = E[G_t | s_t = s]$

• Action Value (Q-function): $Q^\pi(s, a) = E[G_t | s_t = s, a_t = a]$

Where $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ is the return

Handwritten table for Q-values:

	a_0	a_2	a_3	a_1	a_5	
s_1	0.1	0.1	0.3	0.2	0.3	= 1.0
s_2	0.3	0.1	0.2	0.1	0.1	= 1.0
s_3	0.3	0.3	0.2	0.1	0.1	= 1.0



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Value Functions

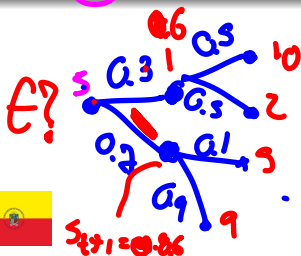
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Where $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ is the **return**.

how good is the future?

cumulative future



$$0.3 (0.3 \times 10 + 0.7 \times 2) \quad 4$$

$$0.7 (0.3 \times 5 + 0.7 \times 9) = V_1 \quad \sim 8$$

$$s_0 - f = 10$$

$$s_1 - f = 13$$

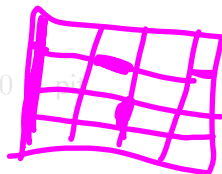
$$s_2 - f = 11$$

$$s_1 - f = 8.6$$



Practical Example: Grid World Navigation

- **Environment:** 4×4 grid with walls and goal
- **States:** Agent's position (x, y)
- **Actions:** Up, Down, Left, Right
- **Rewards:** -1 per step, $+100$ at goal, -100 at pit
- **Goal:** Find shortest path to reward



Practical Example: Grid World Navigation

- **Environment:** 4×4 grid with walls and goal
- **States:** Agent's position (x, y)
- **Actions:** Up, Down, Left, Right

- **Rewards:** $s_1 - 7$, $s_2 - 9-10$
- **Goal:** Find shortest path to reward



$\uparrow - 0.7$
 $\leftarrow - 0.0$
 $\rightarrow - 0.0$
 $\downarrow - 0.3$

π



$\uparrow - 0.2$
 $\leftarrow - 0.8$
 $\rightarrow - 0.0$
 $\downarrow - 0.0$



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constraints

iterations:

QL loop:



Algorithms



Q-Learning Algorithm I

Definition

Q-Learning is a **model-free** reinforcement learning algorithm that learns the **optimal** action-value function $Q^*(s, a)$ directly from interactions with the environment.

Optimize
expected
value

⇓
expectation
maximization

$a_1 \rightarrow$ best action

$$\begin{array}{l} s_1 a \rightarrow s'_1 \rightarrow \epsilon \downarrow \\ \hline a \rightarrow s_1 \rightarrow \epsilon \downarrow \\ a_2 \rightarrow s_3 \rightarrow \epsilon \downarrow \\ a_3 \rightarrow s_2 \rightarrow \epsilon \uparrow \end{array}$$



Q-Learning Algorithm II

Q-Learning Update Rule

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

Handwritten annotations:
 - "immediately" points to r_{t+1}
 - "Future" points to $\gamma \max_a Q(s_{t+1}, a)$
 - s_t, a_t are circled in blue
 - s', a' is written in red and bracketed, pointing to the next state-action pair in the sequence.

- α : Learning rate $[0, 1]$
- γ : Discount factor $[0, 1]$
- **Off-policy**: Learns optimal policy regardless of behavior policy
- **Model-free**: No need to know transition probabilities

Handwritten: Bellman Equation

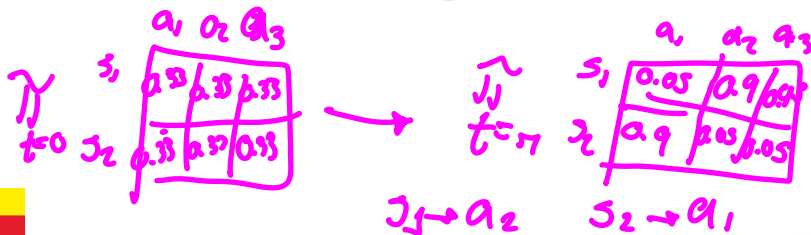


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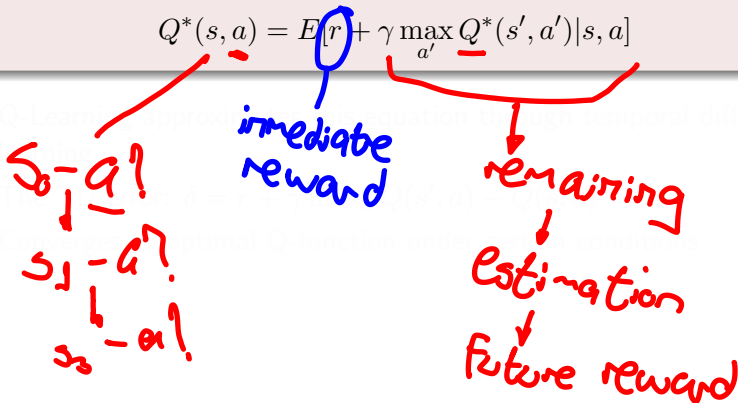


Bellman Equation for Q-Values

~1950-1960

Bellman Optimality Equation

$$Q^*(s, a) = E[r + \gamma \max_{a'} Q^*(s', a') | s, a]$$



Bellman Equation for Q-Values

Bellman Optimality Equation

$$Q^*(s, a) = E[r + \gamma \max_{a'} Q^*(s', a') | s, a]$$

- Q-Learning approximates this equation through temporal difference learning
- The TD error: $\delta = r + \gamma \max_a Q(s', a) - Q(s, a)$
- Converges to optimal Q-function under certain conditions

$$Q(s, a) + \gamma [r + \max_{a'} Q(s', a') - Q(s, a)]$$



Bellman Equation for Q-Values

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$$TD \geq 0 \quad \checkmark$$

$TD < 0 \rightarrow$ wrong decision



Bellman Equation for Q-Values

Bellman Optimality Equation

$$Q^*(s, a) = E[r + \gamma \max_{a'} Q^*(s', a') | s, a]$$

- Q-Learning approximates this equation through **temporal difference** learning
- The **TD error**: $\delta = r + \gamma \max_a Q(s', a) - Q(s, a)$
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Greedy $\rightarrow$ coins
 real! 500? -- 50?



Exploration vs. Exploitation

The Exploration-Exploitation Dilemma

Should the agent choose the **best known action** (exploit) or try **something new** (explore)?

ϵ -Greedy Strategy

- With probability ϵ choose random action (explore)
- With probability $1 - \epsilon$ choose best known action (exploit)
- Often use decaying ϵ over time

exploitation $\rightarrow$ gradient
exploration



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- Often use **decaying** ϵ over time

$\epsilon \rightarrow 0.9 \rightarrow \epsilon \rightarrow 0.5 \rightarrow \epsilon \rightarrow 0.1$

while True:
 random = 0-1
 if random < ϵ :
 $a = \text{random_a}()$
 else:
 $a = \max_a()$
 $\epsilon = \epsilon - 0.01$



Practical Example: Q-Learning in Grid World

- Initialize $Q(s, a)$ arbitrarily (e.g., all zeros)
- For each episode:
 - Initialize state s
 - Repeat until episode ends:
 - Choose action a using ϵ -greedy policy
 - Take action a , observe reward r and new state s'
 - Update Q-value: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_a Q(s', a) - Q(s, a)]$
 - Update state: $s \leftarrow s'$



Practical Example: Q-Learning in Grid World

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$a \rightarrow a_n \rightarrow \text{goal}$
 $s' \rightarrow s_n$



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SARSA Algorithm

SARSA Update Rule

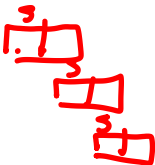
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- **On-policy**: Updates based on the **actual policy** being followed
- **SARSA**: State-Action-Reward-State-Action sequence

- More conservative than Q-Learning
- Better for safety critical applications

Q-Learning: $r + \gamma V[\dots \text{goal}]$

SARSA: $r + \text{penalty (1 step future)}$



SARSA Algorithm

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- More **conservative** than **Q-Learning** → **best action**
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exploration



On-Policy vs. Off-Policy Learning

On-Policy (SARSA):

- Learns about policy currently following } $s \rightarrow s'$
- More conservative
- Accounts for exploration in updates
- Safer in risky environments

Off-Policy (Q-Learning):

- Learns about optimal policy *Future actions*
- More aggressive
- Ignores exploration in updates
- Faster convergence

exploitation
↓
greedy with a click



SARSA Algorithm Mechanics

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 - Update state and action: $s \leftarrow s', a \leftarrow a'$



Practical Example: Cliff Walking Problem

- **Environment:** Agent walks along a cliff edge
- **Cliff:** Large negative reward (-100)
- **Goal:** Reach the end safely
- **SARSA:** Takes safer path (away from cliff)
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Function Approximation in RL

The Problem

- Large or **continuous state spaces**
- Cannot store **Q-values** for every state-action pair
- Need to **generalize** from seen to unseen states

Solution: Function Approximation

Use neural networks to approximate: $Q(s, a) \approx Q(s, a; \theta)$



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Deep Q Network (DQN) Architecture

Key Components

- **Deep Neural Network:** Approximates Q-function
- **Experience Replay:** Stores and samples past experiences
- **Target Network:** Stable Q-targets for training
- **Fixed Q-Targets:** Reduces correlation in updates

Loss Function

$$L(\theta) = E[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]$$



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Experience Replay Mechanism

- **Store:** Experiences (s, a, r, s') in replay **buffer**
- **Sample:** Random minibatch for training
- **Benefits:**
 - Breaks temporal correlations
 - Improves data efficiency
 - Enables batch learning



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DQN for Atari Breakout: Practical Example

Atari Breakout Implementation

- **Input:** Raw pixels ($210 \times 160 \times 3$) $\rightarrow$ preprocessed ($84 \times 84 \times 4$)
- **Network:** CNN with 3 conv layers + 2 dense layers
- **Actions:** 4 discrete actions (NOOP, FIRE, RIGHT, LEFT)

• Preprocessing: Grayscale, resize, frame stacking (4 frames)

• Reward: +1 for each brick destroyed, +5 for clearing level

• Strategy: Agent learns to shoot bricks and move paddle for maximum score

• Performance: Achieves superhuman performance (400 points)



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Policy-Based vs. Value-Based Methods

Value-Based:

- Learn **value functions**
- Derive policy from values
- Discrete action spaces
- Q-Learning, SARSA

Policy-Based:

- Learn **policy directly**
- Parameterized policies
- **Continuous** actions
- Stochastic policies



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Policy Gradient Theorem

Gradient of Expected Return

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}}[\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)]$$

- θ : Policy parameters
- Intuition: Move parameters in direction that increases probability of good actions
- Key insight: We can estimate gradients through sampling



Policy Gradient Theorem

Gradient of Expected Return

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}}[\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)]$$

- $\hat{\theta}$: Policy parameters
- **Intuition:** Move parameters in direction that **increases probability** of good actions
- **Key insight:** We can estimate gradients through **sampling**



REINFORCE Algorithm

REINFORCE Update

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t$$

- Uses **Monte Carlo** return G_t
- High **variance** but unbiased
- Often use **baseline** to reduce variance
- $\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b(s_t))$



REINFORCE Algorithm

REINFORCE Update

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Actor-Critic Architecture

Two Components

- **Actor:** Policy $\pi_{\theta}(a|s)$ — **chooses** actions
- **Critic:** Value function $V_{\phi}(s)$ — **evaluates** actions

Benefits

- Lower variance than REINFORCE
- Online learning (no need to wait for episode end)
- Uses TD error: $\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$



Actor-Critic Architecture

Two Components

- **Actor:** Policy $\pi_{\theta}(a|s)$ — **chooses** actions
- **Critic:** Value function $V_{\phi}(s)$ — **evaluates** actions

Benefits

- **Lower variance** than REINFORCE
- **Online learning** (no need to wait for episode end)
- Uses TD error: $\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$



Proximal Policy Optimization (PPO) Motivation

Problems with Basic Policy Gradients

- Large policy updates can be **destructive**
- Hard to choose appropriate **step size**
- Poor **sample efficiency**

PPO Solution

Constrain policy updates to stay in a **trust region** around current policy



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PPO Clipped Objective

Clipped Surrogate Objective

$$L^{CLIP}(\theta) = E[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

- $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ (probability ratio)
- $\hat{A}_t$ is advantage estimate
- Clips ratio to $[1 - \epsilon, 1 + \epsilon]$
- Prevents **destructively large** policy updates



Practical Example: PPO on LunarLander-v2

LunarLander-v2 Environment

- **State:** Position, velocity, angle, angular velocity, leg contact (8D continuous)
- **Actions:** 4 discrete actions (NOOP, LEFT, MAIN, RIGHT)
- **Goal:** Land safely between flags (+100 to +140 points)

• Reward Shaping: Distance, speed, angle penalties, fuel costs

• PPO Advantages: Stable learning with sparse rewards

• Policy Network: MLP ($8 \rightarrow 128 \rightarrow 128 \rightarrow 4$)

Training: Converges in 1000 – 3000 episodes

Performance: Solves environment (Score > 200)



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- 1 Introduction to Reinforcement Learning Fundamentals
- 2 Advanced Reinforcement Learning Algorithms
- 3 Policy Optimization Methods
- 4 Advanced Policy Methods & Multi-Agent RL**



Trust Region Concept

Trust Region Optimization

- Optimize within a **trusted region** of current policy
- Ensures **monotonic improvement**
- Prevents policy **collapse**

Trust Region Policy Optimization (TRPO) Constraint

$$E[KL(\pi_{\theta_{old}}(\cdot|s), \pi_{\theta}(\cdot|s))] \leq \delta$$



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TRPO Mathematical Formulation

Constrained Optimization Problem

$$\text{maximize} \quad L(\theta) \quad (1)$$

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- Solved using conjugate gradient method
- More principled than PPO but computationally expensive
- Guarantees monotonic improvement



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Natural Policy Gradients

Natural Policy Gradient

- Uses **Fisher information matrix** to scale gradients
 - Accounts for **geometry** of policy space
 - Update: $\theta \leftarrow \theta + \alpha F^{-1} \nabla_{\theta} J(\theta)$
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- More **stable** updates than vanilla policy gradients
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Practical Example: TRPO in Complex Continuous Control

Complex Continuous Control Tasks

- **High-dimensional** state and action spaces
- Requires **precise** control and coordination
- Examples: *robotic manipulation, locomotion*

- TRPO Advantages: Handles complex dynamics and constraints
- Policy Network: Deep neural networks for representation
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Multi-Agent Environments

Challenges

- **Non-stationary** environments
- Other agents are **learning** simultaneously
- **Coordination** and competition
- **Communication** between agents

• **Cooperative:** Agents share common goal

• **Competitive:** Zero-sum games

• **Mixed:** Combination of cooperation and competition



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Nash Equilibrium

A strategy profile where no agent can improve by **unilaterally changing** their strategy.

- Difficult to compute in practice
- May not always exist or be unique
- Agents may not converge to equilibrium
- Alternative: Find correlated equilibria



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Centralized Training, Decentralized Execution

CTDE Paradigm

- **Training:** Use **global information** and centralized critic
- **Execution:** Each agent acts with only **local observations**

- Addresses partial observability
- Enables coordination during training
- Maintains decentralized execution
- Examples: MADDPG, COMA, QMIX



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Communication in Multi-Agent Systems

Communication Protocols

- **Explicit:** Agents exchange messages
- **Implicit:** Agents infer others' intentions from actions

- Communication can enhance coordination and performance
- Challenges: Bandwidth constraints, noise, security
- Research area: Learning to communicate, end-to-end



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/reinforcement-learning>

